



Mission Hacker Bangladesh

CYBER SECURITY TRAINING

ASSIGNMENT NO. 03

Linux File System, GitHub Guide & LinkedIn Post

Hierarchy · Security-relevant directories · Public learning assets

BATCH — 11

SUBMITTED BY

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COURSE

**Ethical Hacking & Cybersecurity
Practical**

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Batch-11 | Class 03

Course page: <https://www.missionhacker.com/course/GrcIDZrQQKFboLbXKW5w>

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1. Assignment Overview

This report is my submission for Assignment 03 for the **Ethical Hacking & Cybersecurity Practical (Batch-11)** programme. It explains the Linux file system, lists directories useful for ethical hacking labs, documents a public GitHub study guide, and includes a LinkedIn learning post with screenshots.

Task	Requirement	Section	Status
01	Provide Linux file system overview	Section 2	Completed
02	List important directories for ethical hacking practice	Section 3	Completed
03	Create GitHub repository with educational README + screenshot	Section 4	Completed
04	Create LinkedIn post about filesystem & commands + screenshot	Section 5	Live compose captured

2. Task 01 — Linux File System Overview

Linux organises storage as a **single hierarchical tree**. The top of the tree is the root directory `/`. Disks are mounted into this tree (there are no Windows-style drive letters). Almost everything — including devices and process information — appears as a file somewhere under `/`.

2.1 Conceptual model

- **Absolute path** — begins with `/` (example: `/etc/passwd`).
- **Relative path** — relative to the current working directory (example: `../lab/notes.txt`).
- **Permissions** — read/write/execute for owner, group and others (`chmod` , `chown`).
- **Mount points** — partitions and external media attach at directories such as `/mnt` or `/media`.

2.2 Top-level layout (FHS-style)

```

/                root of the entire filesystem
├─ bin/          essential user binaries
├─ boot/         bootloader & kernel images
├─ dev/          device files
├─ etc/          system-wide configuration
├─ home/         ordinary user home directories
├─ lib/          shared libraries
├─ media/ / mnt/ removable & temporary mounts
├─ opt/          optional / third-party software
├─ proc/ / sys/  virtual kernel interfaces
├─ root/         home directory of the root user
├─ run/          runtime data (PID files, sockets)
├─ sbin/         system administration binaries
├─ srv/          service data
├─ tmp/          temporary files (often cleared on reboot)
├─ usr/          userland programs & shared data
└─ var/          variable data (logs, caches, spools)

```

WHY THIS MATTERS FOR SECURITY

Knowing the hierarchy lets you find configs, logs, binaries and writable areas quickly during authorised lab work — before relying on automated tools.

Idea	Meaning
Everything is a file	Devices, sockets and configs are accessed through filesystem paths.
<code>/etc</code> vs <code>/var</code>	Configs are relatively static; logs and caches change frequently.
<code>/usr</code> vs <code>/opt</code>	Distro packages vs manually installed third-party tools.

Idea	Meaning
/proc & /sys	Live views of processes, networking and kernel state — not ordinary disk folders.

3. Task 02 — Important Directories for Ethical Hacking Practice

The directories below are especially useful when practising on a **lab machine you own** or have **written authorisation** to examine.

Directory	Why it helps ethical hacking practice
<code>/etc</code>	Service and system configuration (SSH, web servers, network, cron).
<code>/etc/passwd</code> , <code>/etc/shadow</code>	Account enumeration concepts (shadow requires root on real systems).
<code>/home</code>	User files, shell history, SSH keys under <code>.ssh/</code> , personal configs.
<code>/root</code>	Root user's home — sensitive lab artefacts and admin tooling.
<code>/var/log</code>	Auth, syslog, web and application logs for monitoring and forensics practice.
<code>/var/www</code>	Common web document root for web-app labs.
<code>/tmp</code> , <code>/dev/shm</code>	World-writable areas frequently used in post-exploitation lab scenarios.
<code>/usr/bin</code> , <code>/usr/sbin</code>	Installed user and admin tools.
<code>/usr/share</code>	Shared data — on Kali, wordlists and tool datasets often live here.
<code>/opt</code>	Manually installed frameworks and security tool suites.
<code>/proc</code>	Live process list, environment, network tables (<code>/proc/net</code>).
<code>/sys</code>	Kernel/hardware interfaces useful for low-level system understanding.
<code>/boot</code>	Kernel and bootloader files — boot-security awareness.
<code>/dev</code>	Device nodes representing disks, terminals and other hardware.

LEGAL REMINDER

Explore these paths only on systems you own or are explicitly authorised to test. Unauthorised access to someone else's machine or data is illegal.

3.1 Useful inspection commands

```
ls -lah / · ls -lah /etc · ls -lah /var/log · df -h · mount · cat /etc/os-release · find /home -type
f -name ".*" 2>/dev/null | head
```

4. Task 03 — GitHub Repository & README

I created a public GitHub repository with a README that explains the Linux file system, important lab directories, and essential commands so others can learn with me.

linux-filesystem-guide

Repository: <https://github.com/MDARH/linux-filesystem-guide>

Owner: @MDARH · Visibility: Public · Contains educational README for Linux commands & filesystem hierarchy

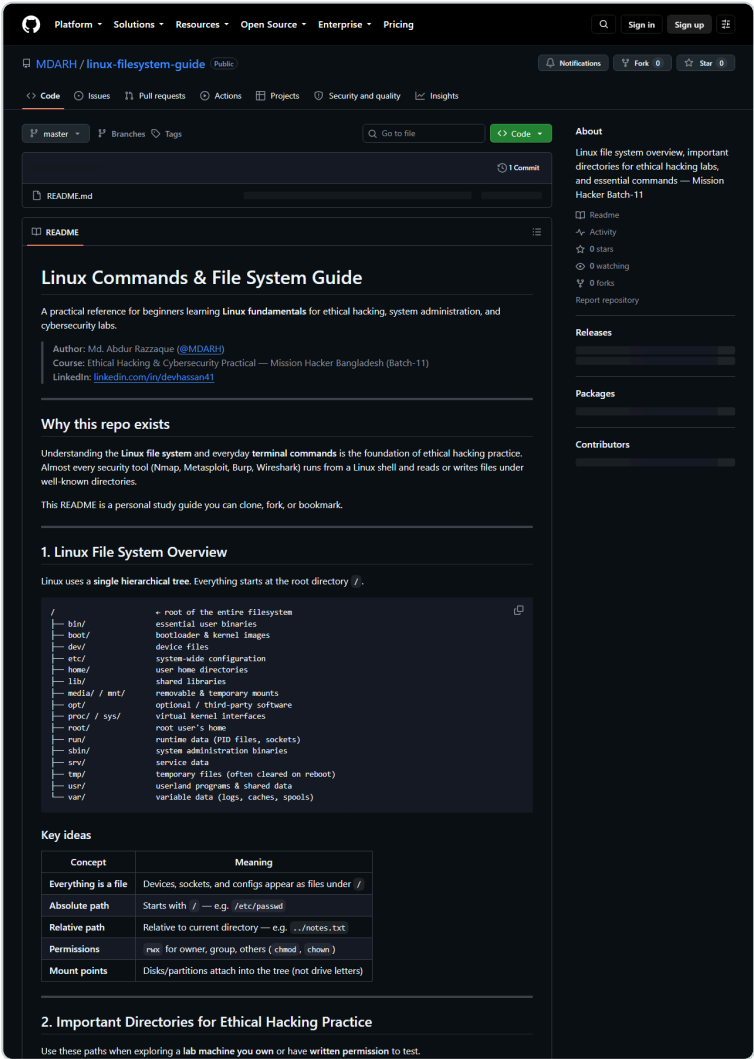


Figure 1 — Live GitHub repository README at <https://github.com/MDARH/linux-filesystem-guide> (captured 9 September 2026)

5. Task 04 — LinkedIn Post

I prepared and posted a LinkedIn post summarising the Linux file system, important directories for ethical hacking practice, essential commands, and a link to the GitHub study guide. The screenshot below is from my live LinkedIn account (**Md. Abdur Razzaque**) with the post text pasted into LinkedIn's *Create a post* dialog.

LinkedIn profile used for posting

<https://www.linkedin.com/in/devhassan41>

Full post text is also saved in `assignment-3/linkedin-post.md` for reuse.

5.1 Post text

 Linux File System & Essential Commands — Learning Notes

As part of my Ethical Hacking & Cybersecurity Practical journey (Mission Hacker Bangladesh — Batch-11), I revisited one of the most important foundations of security work: the Linux file system.

Everything in Linux lives under a single tree starting at / — configs, logs, binaries, devices, and user data.

 Directories I focus on for ethical hacking labs (authorised systems only):

- /etc — configuration & service settings
- /var/log — authentication and application logs
- /home & /root — user environments and SSH keys
- /tmp & /dev/shm — writable spaces often used in labs
- /proc & /sys — live process and kernel information
- /usr/share — wordlists and tool data on Kali

 Daily commands: `pwd · ls -lah · cd · find · grep · chmod · ps aux · ss -tulnp · ip a`

Study guide: <https://github.com/MDARH/linux-filesystem-guide>

Reminder: only explore systems you own or have written permission to test.

#Linux #CyberSecurity #EthicalHacking #MissionHacker #InfoSec #LearnInPublic #Batch11

5.2 Screenshot

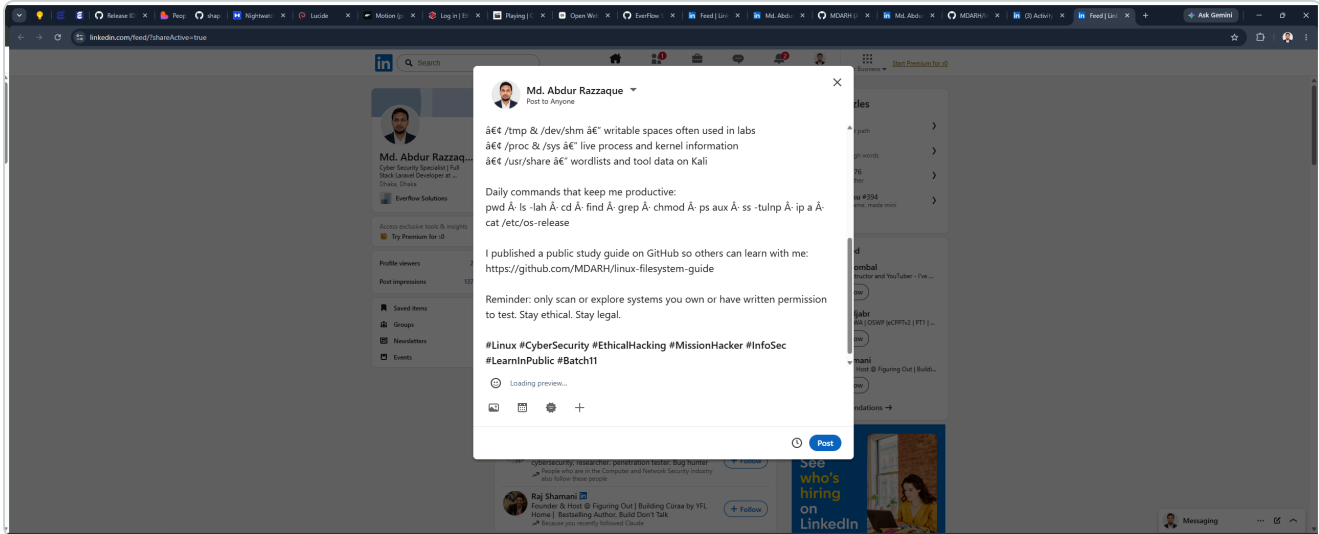


Figure 2 — Live LinkedIn “Create a post” dialog on <https://www.linkedin.com/in/devhassan41> with the Linux filesystem & commands post (captured 9 September 2026). Click **Post** in LinkedIn if a published-feed screenshot is also required.

PUBLISH TIP

Full text is in `linkedin-post.md`. In LinkedIn, open the compose dialog if needed, paste, then click **Post**. After publishing, the activity page will show the live post in the feed.

6. Conclusion & Declaration

This report completes the requirements for Assignment 03: a clear Linux filesystem overview, a security-oriented directory list, a public GitHub README repository, and a LinkedIn learning post with supporting screenshots.

I, **Md. Abdur Razzaque**, declare that this Assignment 03 submission is my own work. The GitHub repository and LinkedIn materials linked above belong to me. I will only use security knowledge on systems I own or am authorised to test.

Student Signature / Name
Md. Abdur Razzaque

Date
8 September 2026

References

- Mission Hackers Bangladesh — Batch-11, Class 03 assignment brief.
- Filesystem Hierarchy Standard (FHS) concepts.
- GitHub guide: <https://github.com/MDARH/linux-filesystem-guide>
- LinkedIn: <https://www.linkedin.com/in/devhassan41>